

# Raphael Z. Stone

7208 Lamplight Lane, Austin, TX 78731  
raphaelzstone@gmail.com, 512-810-3881

## EDUCATION

**The University of Chicago, Chicago, IL**

*Expected June 2024*

BS in Molecular Engineering

Minor in Sustainable Energy and Water Resources

GPA: 3.43/4.0

**The Liberal Arts and Science Academy (LASA), Austin, TX**

*Graduated June 2020*

## AWARDS AND HONORS

Quad Research Scholar, supported by NK Cheung Chemistry Research Fellowship, October 2023

Quad Summer Research Scholar, June 2023

Quad Research Scholar, January 2023

Jeff Metcalf Research Grant, June 2022

## RESEARCH AND WORK EXPERIENCE

[Prof. Rachel Kurchin's Group](#) at Carnegie Mellon University, Pittsburgh, PA

**Ph.D. Student**, August 2024 – Present

- Leading project on efficient modeling of glassy materials using a DFT ensembling technique. I perform DFT using the TRACE supercomputer, and do data analysis to produce simulated XRD,  $g(r)$ , bulk modulus, and other materials properties.
- Working on a collaboration to increase materials discovery rate using DFT, MD, and ML. I perform DFT and MD to downselect materials phase-space and create training data for ML models to learn from.
- Leading project on differentiable DFT in JuliaMolSim ecosystem for point defect calculations.

[Prof. Chong Liu's Group](#) at the University of Chicago, Electrochemical Separation Subgroup, Chicago, IL

**Research Assistant**, September 2021 – July 2024

- Leading project on Li/Na selectivity and ion-exchange kinetics with the goal of electrochemical lithium extraction from brines and seawater. Synthesized, etched, and exfoliated  $\text{LiCoO}_2$  (LCO) for the study of particle size effects.
- Electrochemically probed LCO Li/Na extraction selectivity, capacity, lifetime using chronopotentiometry (CP).
- Independently characterized synthesized particles using XRD, SEM, and ICP-MS, analyzing particle facet ratios, particle crystallinity, and composition of tested material.
- Synthesized and tested  $\text{LiCoO}_2$ ,  $\text{LiNi}_{1/3}\text{Mn}_{1/3}\text{Co}_{1/3}\text{O}_2$ ,  $\text{LiTiO}_2$ , and numerous other layered oxide materials.

[SparkCognition](#), Austin, TX

**Intern**, June – September 2021 and June – September 2022

- Researched and developed in Python on machine learning for seismology. Applied inverse problems to deep neural networks with the goal of faster seismic imaging.
- Developed and ran image comparison tests to analyze and improve machine learning algorithm outputs. Built imaging and visualization tools for 3-D analysis of seismic data, enabling effective tuning of models.
- Wrote and ran scripts on the Longhorn and Stampede supercomputers at the Pickle Research Center, running machine learning algorithms on hundreds of datasets.

[Prof. Yair Ein-Eli's Lab](#) at Technion – Israel Institute of Technology, Haifa, Israel

**Research Intern**, June – September 2019

- Performed chronopotentiometry (CP) on carbon nanotubes to electroplate copper, analyzing conductivity
- Used SEM to characterize results from tests changing current, mixing, and solution properties.

## PUBLICATIONS

- Grant Hill, **Raphael Stone**, Yu Han, Jiadong Liu, Siqi Zou, Hua Zhou. [Asymmetric pathways for lithium extraction and recovery based on the two-phase equilibrium of layered oxides](#) *Under Review, July 30, 2024.*
- Grant T. Hill, **Raphael Stone**, Gangbin Yan, Chong Liu. **Nano- $\text{LiCoO}_2$  Electrochemical Li extraction in brines** (Tentative title). *In Preparation, 2024.*

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## PRESENTATIONS

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- MS&T Technical Meeting, Poster Presentation. [Downselection of Crystalline Ensembles for Efficient First-Principles Modeling of Glassy Materials](#). October, 2025.
- University of Chicago Undergraduate Research Symposium, Poster Presentation. [Li Extraction from Unconventional Water Sources via Nano-LiCoO<sub>2</sub>](#). April 2024.
- SparkCognition, Slideshow Presentation. **Denoising of Langevin Dynamic Crosscut in a 3-D Seismic Image Volume**. September 2022.
- SparkCognition, Slideshow Presentation. **2-D Langevin Denoising and Inverse Problem Solving for Seismic Imaging**. September 2021.
- Technion – Israel Institute of Technology, Poster Presentation. **Towards Lightweight Electronics: The Development of Copper Coated Carbon Fabric via Electrochemical Deposition**. September 2019.

## CAMPUS LEADERSHIP

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### **Carnegie Mellon University Crew, Pittsburgh, PA: Assistant Coach (2024-Present)**

- Coach, mentor, and help run the Carnegie Mellon Crew team.
- Develop injury prevention training and awareness, and teach about water safety and cleanliness.

### **The University of Chicago Crew, Chicago, IL: President (2023) and Internal Vice-President (2022)**

- Fundraised, purchased boats, repaired boats, managing just shy of \$100k in club funds and \$200k in club equipment. Also hired coaches, and rented boathouse space from the Chicago Park District.
- Restarted a rowing outreach program for senior citizens in the Chicago area with two neighboring boat clubs, generating over \$40k in revenue for the Park District.
- Helped build a case to rezone a branch of the Chicago River to have maximum environmental protections.

### **The University of Chicago Bridge Club, Chicago, IL: President (March 2021 – March 2023)**

- Built the club from 8 people to over 50 people, leading the club to be the National Champions
- Taught bridge and organized travel for over 100 students, handling \$20k in funds for the club.

### **Chamberlin House at The University of Chicago, Chicago, IL: Vice-President (September 2020 – June 2022)**

- Organized social and competitive events. Handled funds, led intramural teams, and built competitive groups for intra-house competitions.

## SKILLS

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**Laboratory:** Synthesis of battery materials, electrochemical testing with EC-Lab, XRD, NMR, SEM, and ICP-MS.

**Coding:** Python, Java, SLURM, LaTeX, Excel, and R.

**Languages:** Conversational in Hebrew, Beginner in Spanish

## EXTRACURRICULARS AND HOBBIES

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### **Cello (2005 – Present)**

- David L. Fulton Prize for Orchestral Excellence with The U. of Chicago Symphony Orchestra (2024)

### **Rowing (2016 – Present)**

- American Collegiate Rowing Championships: Silver in V2- (2023), Bronze in V2- (2022).
- National Youth Rowing Championship: Silver in Lightweight 4+ (2020), Competed in V2- (2019).

**Hobbies:** Go, Bridge, Chamber Music, Cooking, and Cycling and Unicycling.

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## REFERENCES

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Prof. Rachel Kurchin (Ph.D. Research Advisor)  
Assistant Research Professor, Materials Science and Engineering  
Carnegie Mellon University  
rkurchin@andrew.cmu.edu

Prof. Xioaying Liu (Instructing Professor)  
Associate Senior Instructional Professor & Laboratory Director for Molecular Engineering  
Pritzker School of Molecular Engineering  
The University of Chicago  
liuxy@uchicago.edu, 773-702-5884

Dr. Bruce Porter (Internship Advisor)  
Chief Science Officer  
SparkCognition  
bporter@sparkcognition.com, 512-791-7980

Prof. Chong Liu (Undergraduate Research Advisor)  
Neubauer Family Assistant Professor  
Pritzker School of Molecular Engineering  
The University of Chicago  
chongliu@uchicago.edu, 773-834-5574